

### **Implemented Code (C++):** (Convex Hull: Graham's Scan Algorithm)

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
using vec_p=vector<pair<double,double>>;
using pr=pair<double,double>;
#define A_sort(x,n,y) sort(x.begin()+n,x.end(),y)

double p=1e10,q=1e10;
ll step;
bool compare(pr A,pr B)
{
    double angleA=atan2(A.second-q,A.first-p);
    double angleB=atan2(B.second-q,B.first-p);
    return angleA<angleB;
}
double cross_p(pr o,pr a,pr b)
{
    step++;
    return (a.first-o.first)*(b.second-o.second)-(a.second-o.second)*(b.first-o.first);
}
// using graham scan algorithm
vec_p Grahams_scan(vec_p &points)
{
    int n=points.size();
    vec_p convex_hull;
    for(int i=0;i<3;i++)
    {
        step++;
        convex_hull.push_back(points[i]);
    }
    step++;
    for(int i=3;i<n;i++)
    {
        step++;
        while(convex_hull.size()>1 &&
            cross_p(convex_hull[convex_hull.size()-2],convex_hull.back(),points[i])<=0)
        {
            step++;
            convex_hull.pop_back();
        }
        convex_hull.push_back(points[i]);step++;
    }
    step++;
    convex_hull.push_back(points[0]);step++;
    return convex_hull;
}
```

```

}
int main()
{
    vector<int>value={1000,2000,5000,8000,10000,12000,15000};
    ofstream op;
    op.open("polygon.txt");
    op<<"Total Points--->Total Steps--->Total Vertices\n";
    for(int i=0;i<7;i++)
    {
        ifstream ip;
        ip.open("hull.txt");
        vec_p points;
        step=0;
        double x,y;
        int pp,j=0;
        while(ip>>x>>y)
        {
            points.push_back({x,y});
            if(q>y)
            {
                p=x; q=y; pp=j;
            }
            j++;
            if(j==value[i]){break;}
        }
        ip.close();
        swap(points[0],points[pp]);step++;
        A_sort(points,1,compare);step+=(value[i]-1)*log2(value[i]-1);
        vec_p convex_hull=Grahams_scan(points);
        op<<value[i]<<"----->"<<step<<"----->"<<convex_hull.size()-1<<"\n";
    }
}

```

Output:

| Total Points | Total Step Count | Number of Vertices |
|--------------|------------------|--------------------|
| 1000         | 14920            | 16                 |
| 2000         | 31877            | 20                 |
| 5000         | 86388            | 17                 |
| 8000         | 143673           | 18                 |
| 10000        | 182822           | 19                 |
| 12000        | 222555           | 18                 |
| 15000        | 283032           | 20                 |

